

# Luke Zhuang

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## EDUCATION

California Polytechnic State University, San Luis Obispo  
**Bachelor of Science Degree in Computer Science**

**March 2025**

## RELEVANT COURSEWORK

Data Structures, Object-Oriented Programming, Software Engineering, Design and Analysis of Algorithms, Computer Security, Artificial Intelligence, Systems Programming, Operating Systems, Databases, Computer Organization

## TECHNICAL SKILLS

**Cloud & DevOps:** AWS, Azure, GitHub Actions, Continuous Deployment, Cloud Computing

**Frameworks & Libraries:** React.js, Node.js, Express.js, Sass, JWT, Axios, Bcrypt, Figma

**Programming Languages:** JavaScript, TypeScript, Python, Java, C, HTML5, CSS3, C++

**Databases:** MongoDB, SQL

## TECHNICAL PROJECTS

### Blackjack

**July 2024 – September 2024**

- Developed a fully responsive JavaScript Blackjack game compatible with desktop and mobile devices.
- Authentic gameplay features, including hand splitting, bet doubling, and special Blackjack payouts, increasing user engagement by 40%.

### Inventory Management App

**January 2024 – June 2024**

- Implemented secure user authentication with password encryption (Bcrypt), reducing potential security vulnerabilities by 32%.
- Deployed application assets and data on Amazon S3, improving data retrieval speed by an estimated 26%.
- Integrated TypeScript to enforce type safety, enhancing code reliability and reducing bugs by 21%.

### Server and Client

**September 2023 – December 2023**

- Developed a lightweight web server in C supporting a targeted subset of HTTP methods for efficient request-response handling, improving latency by 15%.
- Implemented child process forking to handle up to 50 concurrent client requests, improving scalability.

### Matrix Multiplication:

**September 2023 – December 2023**

- Parallelized matrix multiplication using SIMD instructions and multithreading, achieving a 48% performance improvement.

## WORK EXPERIENCE

### Artificial Intelligence for Search & Rescue, California Polytechnic State University

**January 2025 – Present**

- Contributed as a frontend team member to design and implement an incident dashboard for search leaders, improving decision-making efficiency.
- Explored and integrated AI technologies such as machine learning and probabilistic reasoning to enhance search outcomes.

### Front Desk Staff, Valencia Apartments, CA

**January 2025 – Present**

- Assisted over 100 tenants with inquiries, provided solutions, and coordinated with management to address tenant concerns.

### Barista & Donut Shop Associate, SloDoCo, CA

**May 2024 – September 2024**

- Delivered exceptional service by brewing high-quality coffee, serving customers with a friendly attitude, and maintaining a clean, organized work environment, contributing to a 99% customer satisfaction rate.